SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I M.Ramprasadh/RA2513055 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

M.Ramprasadh
Signature of the student:

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
    a. Show differences and similarities
        i. Map how each affects the Data structure and Security level
        ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

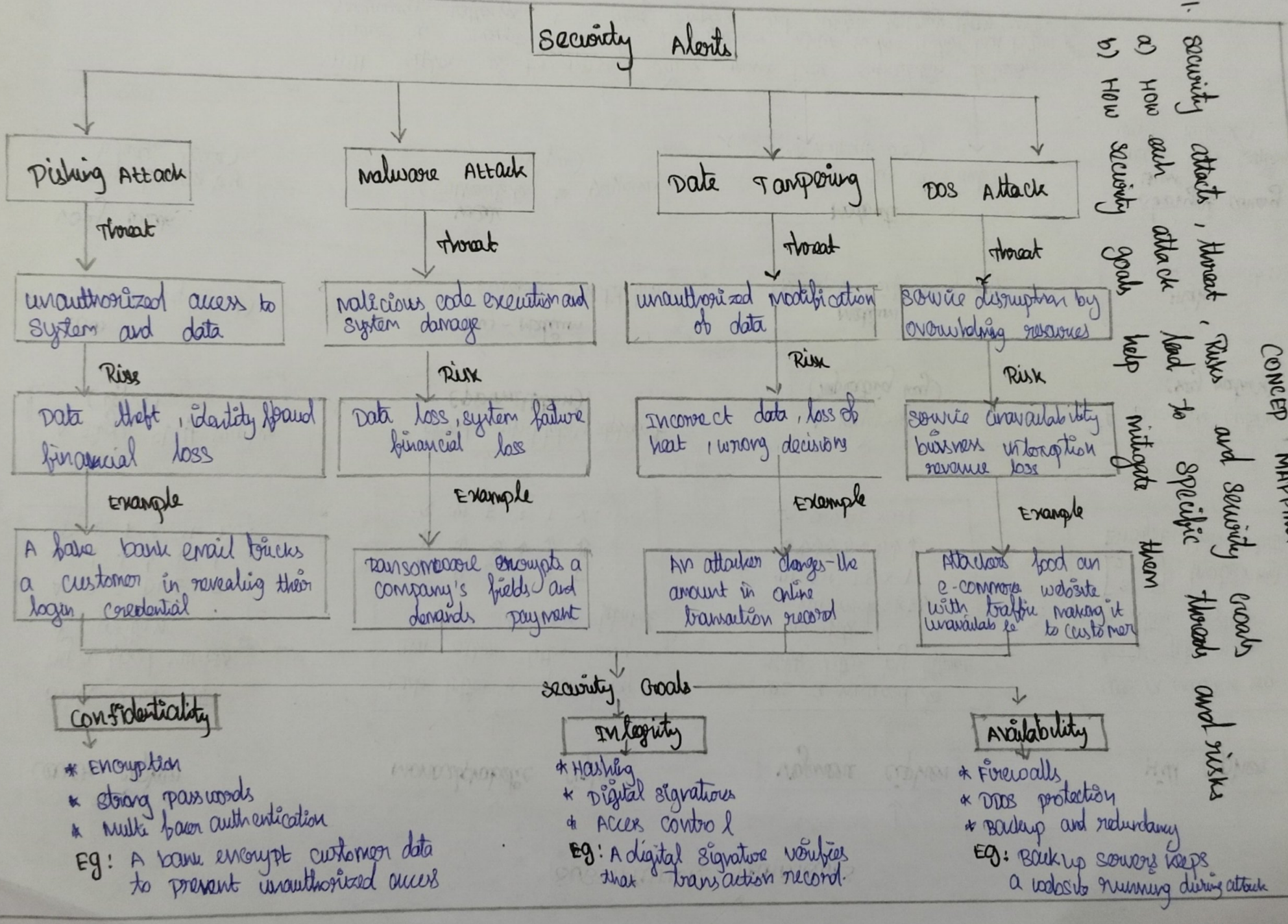
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Hiding

1. Create a concept map connecting security attacks, threats, risks and security goals
   a) How each attack lead to specific threats and risks
   b) How security goals help mitigate them
   c)

CONCEPT MAPPING



SUBSTITUTION TECHNIQUES

Caesar cipher

Shift each letter by a fixed number of position

A B C D I E F ...
↓ ↓ ↓ ↓ ↓ ↓
D E F G H I ...

A single shift number (eg. 3)

Low (simple to implement)

Very weak (easily broken by brute force)

Monoalphabetic cipher

Each letter is replaced by any other letter using a fixed substitution alphabet

A B C D E F ...
↓ ↓ ↓ ↓ ↓ ↓
Q W E R T Y ...

Fixed substitution alphabet (permutation)

Low - Medium (Simple - substitution)

Weak (vulnerable to frequency analysis)

Vigenere cipher

Use a keyword to shift letter by different amounts

keyword : KEY
PLAINTEXT
↓↓↓↓↓↓↓↓↓
ZODVRCXBT

A keyword (repeating key)

Medium (multiple shift using key)

Moderate (Harder than caesar and monoalphabetic)

Hill cipher

Use a matrix key to encrypt group of letters together

$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$ (matrix key)

Encrypt block of letters

A invertible matrix (key matrix)

High (matrix operations)

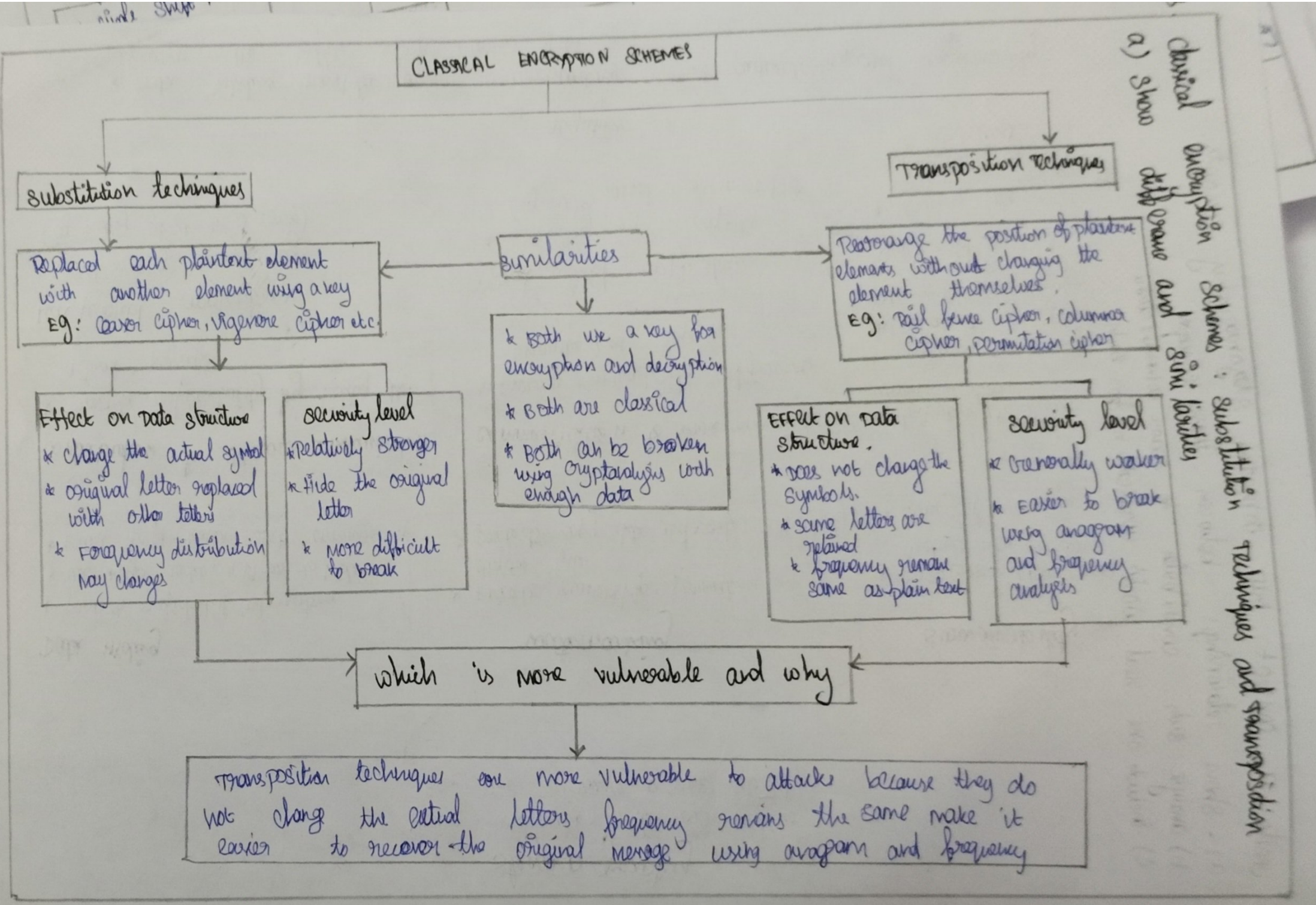
Strongest among these (Hides letter frequency using matrix)

Convert Map from Substitution techniques

Hill cipher is the most secure among these substitution techniques because it encrypt block of using matrix which hides individual letter frequency patterns. It is more complex and harder to break than caesar.

Classical encryption schemes : substitution techniques and transposition techniques

a) Show differences and similarities



4. Create a concept map connecting steganography concepts.
a). Show relationships between these concepts
b) analyze each contributes to secure communication
c) include one real world application for each

Steganography

Hiding information inside cover medium
so that presence is not detected

Data Hiding

* protects confidential information
* use images, audio or video to hide.
* Extra point: makes secret communication

Contribution to secure communication

* provides confidentiality by hiding the actual message

Real world applications

* secret message sharing in images
(eg. hiding text in a photo)

Watermarking

* Embeds ownership information in digital files
* protects copyrights and verifies authenticity

Communication to secure communication

* Ensures authenticity and prevents unauthorized copying

Real world applications

* Embedding logo in images
(eg. TV channel water mark)

Survivability

* Ensures hidden remains include after processing
* main goal: robustness

Contribution to secure communication

* maintains availability and reliability of hidden data even after modification

Real world applications

* Hidden data in image that survive compression. (eg: Social media)

Together

* Data hiding provides secrecy, watermarking proves authenticity and survivability